

D427

Data Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A marketing analyst at a regional bakery chain runs `SELECT ... FROM Customers c LEFT JOIN Orders o ON c.id = o.customer_id`. Customers has 500 rows; Orders has 1,200 rows placed by 420 different customers. The query runs in 0.3 seconds. How many customers appear with NULL in the order columns?

- A. 0
- B. 80
- C. 420
- D. 700

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An INNER JOIN between Customers and Orders returns fewer distinct customers than the Customers table holds. Which customers are missing?

- A. Customers with no matching orders.
- B. Orders with NULL fields.
- C. Duplicate customer records.
- D. Customers who have orders.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A sales manager wants regions whose total sales exceed \$1,000,000. The Sales table has 2 million rows and is refreshed at midnight. After GROUP BY region, which clause filters on the total?

- A. ORDER BY SUM(amount) DESC
- B. HAVING SUM(amount) > 1000000
- C. WHERE amount > 1000000
- D. WHERE SUM(amount) > 1000000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A query must exclude cancelled orders before totals are computed per customer. Where does status <> 'cancelled' belong?

- A.** In the ORDER BY clause.
- B.** In the WHERE clause.
- C.** In the GROUP BY clause.
- D.** In the HAVING clause.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

The Employees table has 250 rows, and 40 of them have NULL in manager_id. HR uses the table for payroll in three states. What does SELECT COUNT(manager_id) FROM Employees return?

- A.** 250
- B.** 0, because any NULL makes COUNT return nothing.
- C.** 210
- D.** 40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

In MySQL with ONLY_FULL_GROUP_BY on, SELECT dept, name, COUNT(*) FROM Employees GROUP BY dept; fails. Why?

- A.** Text columns can never be grouped.
- B.** GROUP BY must come before FROM.
- C.** COUNT(*) must name a column.
- D.** name is not in GROUP BY and is not inside an aggregate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst for a 40-store pet supply chain runs `SELECT * FROM Customers WHERE phone = NULL`; to find customers with no phone. It returns zero rows, though about 300 customers have no phone on file. What is the fix?

- A. `WHERE phone IS NULL`
- B. `WHERE phone <> NULL`
- C. `WHERE phone = ''`
- D. `WHERE store_id <= 40 AND phone = NULL`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A buyer wants every product priced above the average price. The catalog has 3,500 SKUs from 60 suppliers. Which query works?

- A. `SELECT name FROM Products HAVING price > AVG(price);`
- B. `SELECT supplier, AVG(price) FROM Products GROUP BY supplier;`
- C. `SELECT name FROM Products WHERE price > AVG(price);`
- D. `SELECT name FROM Products WHERE price > (SELECT AVG(price) FROM Products);`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A DBA at a credit union runs `UPDATE Accounts SET status = 'frozen'`; meaning to freeze only account 55123. The table has 48,000 rows and autocommit is on. What is the result?

- A. All 48,000 accounts are frozen and the change is committed.
- B. Only account 55123 is frozen.
- C. The statement errors because `WHERE` is required.
- D. Nothing changes until `COMMIT` is typed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A `DELETE` removed 12 of the 5,000 rows in `Orders`. What must the statement have included?

- A. A `GROUP BY` clause.
- B. `TRUNCATE` instead of `DELETE`.
- C. A `WHERE` clause that matched those 12 rows.
- D. No `WHERE` clause, since `DELETE` only removes changed rows.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Orders.customer_id references Customers.id with ON DELETE CASCADE. The customer service team works out of Manila. A customer who has 7 orders is deleted. What happens?

- A. The orders stay with customer_id set to NULL.
- B. The orders stay, pointing to a missing customer.
- C. The customer and all 7 orders are deleted.
- D. The delete is blocked because orders exist.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Lookups on Orders by customer_id take 9 seconds on a 20-million-row table, and about 50 new orders arrive each second. The team just upgraded to MySQL 8. What is the best step, and what is its cost?

- A. Add the index; it slows reads but speeds writes.
- B. Create a view over Orders filtered by customer.
- C. Index every column so any query is fast.
- D. Create an index on customer_id; reads get faster, but inserts get slightly slower and use more storage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A marketing analyst ~~at a regional bakery chain~~ runs `SELECT ... FROM Customers c LEFT JOIN Orders o ON c.id = o.customer_id`. Customers has 500 rows; Orders has 1,200 rows placed by 420 different customers. ~~The query runs in 0.3 seconds.~~ How many customers appear with NULL in the order columns?

Target word: unmatched left rows **Decides:** LEFT JOIN keeps customers with no orders, padded with NULLs.

A	Confusion set	That is INNER JOIN behavior.
B	Correct	500 customers – 420 with orders = 80 with none.
C	Wrong stem number	Customers who do have orders.
D	Inverted operation	1,200 – 500 mixes order rows with customer rows.

2. Answer A Concept 1 • transfer

An INNER JOIN between Customers and Orders returns fewer distinct customers than the Customers table holds. Which customers are missing?

Target word: reverse: which rows an INNER JOIN drops **Decides:** INNER JOIN keeps only matches.

A	Correct	Rows with no match on the join key are dropped.
B	Adjacent category	The question is about customers, and NULL non-key fields do not drop rows.
C	Confusion set	Joins do not remove duplicates.
D	Inverted operation	Those are exactly the ones kept.

3. Answer B Concept 2

A sales manager wants regions whose total sales exceed \$1,000,000. ~~The Sales table has 2 million rows and is refreshed at midnight.~~ After `GROUP BY region`, which clause filters on the total?

Target word: filter on an aggregate **Decides:** The condition is on SUM, so it applies to groups.

A	Adjacent category	Sorts groups but removes none.
B	Correct	HAVING filters groups after aggregation.
C	Dropped element	Filters single sales, not regional totals.
D	Confusion set	WHERE runs before grouping and cannot use aggregates.

4. Answer B Concept 2 • transfer

A query must exclude cancelled orders before totals are computed per customer. Where does status `<> 'cancelled'` belong?

Target word: row filter before grouping **Decides:** The condition is on individual rows.

A	Confusion set	ORDER BY only sorts.
B	Correct	WHERE removes rows before they are grouped and summed.
C	Adjacent category	GROUP BY defines groups, it does not filter.
D	Inverted operation	HAVING filters groups after totals already include cancelled rows.

5. Answer C Concept 3

The Employees table has 250 rows, and 40 of them have NULL in manager_id. ~~HR uses the table for payroll in three states.~~ What does SELECT COUNT(manager_id) FROM Employees return?

Target word: COUNT on a column with NULLs **Decides:** COUNT(column) skips NULLs.

A	Confusion set	That is COUNT(*).
B	Absolute language	COUNT ignores NULLs; it does not fail on them.
C	Correct	250 – 40 NULLs = 210.
D	Inverted operation	Counts the NULLs instead of the non-NULLs.

6. Answer D Concept 3 • transfer

In MySQL with ONLY_FULL_GROUP_BY on, SELECT dept, name, COUNT(*) FROM Employees GROUP BY dept; fails. Why?

Target word: why the grouped query fails **Decides:** name is neither grouped nor aggregated.

A	Absolute language	dept could be text and still be grouped.
B	Inverted operation	The clause order is SELECT, FROM, WHERE, GROUP BY.
C	Confusion set	COUNT(*) is valid and counts rows.
D	Correct	Each dept has many names; the engine cannot pick one.

7. Answer A Concept 4

An analyst ~~for a 40-store pet supply chain~~ runs SELECT * FROM Customers WHERE phone = NULL; to find customers with no phone. It returns zero rows, though about 300 customers have no phone on file. What is the fix?

Target word: correct NULL test **Decides:** = NULL is never true.

A	Correct	IS NULL is the only correct test for a missing value.
B	Inverted operation	Any comparison with NULL is unknown, so this also returns nothing.
C	Confusion set	An empty string is a value, not NULL.
D	Camouflage	Adds the store count but keeps the broken test.

8. Answer D Concept 5

A buyer wants every product priced above the average price. ~~The catalog has 3,500 SKUs from 60 suppliers.~~ Which query works?

Target word: compare rows to an aggregate **Decides:** The average must come from a subquery.

A	Confusion set	Without grouping this does not compare each row to the overall average.
B	Camouflage	Follows the supplier detail but lists averages, not products.
C	Dropped element	Aggregates cannot appear directly in WHERE.
D	Correct	The scalar subquery returns one average to compare against.

9. Answer A Concept 6

A DBA at a credit union runs UPDATE Accounts SET status = 'frozen'; meaning to freeze only account 55123. The table has 48,000 rows and autocommit is on. What is the result?

Target word: scope of an UPDATE with no WHERE **Decides:** No WHERE clause.

A	Correct	UPDATE without WHERE touches every row; autocommit saves it at once.
B	Dropped element	The database cannot see intent; the WHERE clause is missing.
C	Confusion set	WHERE is optional in standard SQL.
D	Adjacent category	True only with autocommit off.

10. Answer C Concept 6 • transfer

A DELETE removed 12 of the 5,000 rows in Orders. What must the statement have included?

Target word: reverse: result to clause **Decides:** Only a WHERE clause limits which rows are deleted.

A	Confusion set	GROUP BY does not apply to DELETE.
B	Adjacent category	TRUNCATE empties the whole table.
C	Correct	WHERE sets the scope of a DELETE.
D	Inverted operation	Without WHERE all 5,000 rows go.

11. Answer C Concept 7

Orders.customer_id references Customers.id with ON DELETE CASCADE. ~~The customer service team works out of Manila.~~ A customer who has 7 orders is deleted. What happens?

Target word: effect of CASCADE **Decides:** CASCADE removes child rows with the parent.

A	Adjacent category	That is SET NULL.
B	Dropped element	Ignores the foreign key entirely.
C	Correct	CASCADE carries the delete to child rows.
D	Confusion set	That is RESTRICT or NO ACTION.

12. Answer D Concept 8

Lookups on Orders by customer_id take 9 seconds on a 20-million-row table, and about 50 new orders arrive each second. ~~The team just upgraded to MySQL 8.~~ What is the best step, and what is its cost?

Target word: index and its trade-off **Decides:** Frequent filtering on one column of a large table.

A	Inverted operation	Reverses the trade-off.
B	Confusion set	A standard view stores no data and does not speed the lookup.
C	Absolute language	Each index slows writes; at 50 inserts per second that cost adds up.
D	Correct	Each insert must also update the index.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	INNER vs. LEFT JOIN	INNER JOIN keeps only matching rows; LEFT JOIN keeps every left-table row and fills NULLs where the right table has no match.	Expecting unmatched rows in an INNER JOIN, or none in a LEFT JOIN.	“Every customer, including those with no orders.”
2	WHERE vs. HAVING	WHERE filters rows before grouping; HAVING filters groups after aggregation.	Putting an aggregate in WHERE.	A condition on SUM, COUNT, or AVG.
3	Aggregates and GROUP BY	COUNT(column) skips NULLs while COUNT(*) counts rows; every non-aggregated SELECT column must appear in GROUP BY.	Treating COUNT(col) like COUNT(*).	A column with NULLs and a COUNT.
4	Testing for NULL	NULL is unknown, so = NULL is never true; use IS NULL or IS NOT NULL.	Writing = NULL or confusing NULL with an empty string.	“Missing” values and zero rows returned.
5	Subqueries	Compare a row to an aggregate with a subquery, such as WHERE price > (SELECT AVG(price) ...).	Using an aggregate directly in WHERE.	“Above average,” “more than the typical.”
6	UPDATE and DELETE scope	Without a WHERE clause, UPDATE and DELETE affect every row in the table.	Assuming the database targets only the intended row.	An UPDATE or DELETE with no WHERE.
7	Foreign key actions	ON DELETE CASCADE deletes child rows; RESTRICT or NO ACTION blocks the delete; SET NULL clears the child’s key.	Mixing up which action deletes, blocks, or nulls.	An ON DELETE clause and a parent delete.
8	Indexes and their trade-off	An index on a filtered or joined column speeds reads but adds storage and slows inserts and updates.	Indexing every column, or expecting a view to speed a query.	Slow lookups on a large table by one column.

Course scope (public description): Applications - Data Management - Applications covers conceptual data modeling and introduces MySQL. Students will learn how to create simple to complex SELECT queries, including subqueries and joins, and how to use SQL to update and delete data. Topics covered in this course include exposure to MySQL; creating and modifying databases, tables, views, foreign keys and primary keys (FKs and PKs), and indexes; populating tables; and developing simple Select-From-Where (SFW) queries to complex 3+ table join queries. The following course is a prerequisite: Data Management - Foundations.